

ClearCrest: A Quantum Safe Peer to Peer Electronic Cash System

Vipul aka findance

Abstract. ClearCrest is a proposed peer to peer electronic cash protocol designed for a migration era in which public key signatures must remain secure against future quantum adversaries. It retains proof of work, an unspent-output ledger, fixed supply issuance, and direct value transfer, while replacing vulnerable signature assumptions with versioned post-quantum signature providers. The protocol defines canonical transaction encoding, SHA3 based commitments, explicit amount bounds, deterministic difficulty windows, native channel commitments, hash time locked contracts, compact relay, and compact filters. This paper specifies the intended consensus rules and security boundaries. It does not claim a production launch until the selected post-quantum providers, network implementation, and operational controls have completed independent review.

1. Introduction

A public payment system must let two willing parties exchange value without assigning transaction ordering, money creation, or custody to a central operator. A distributed timestamp ledger addresses the ordering problem when nodes independently validate transactions and extend the chain with proof of work. The long-term safety of that system also depends on the durability of its public key signatures. A sufficiently capable quantum computer could undermine classical discrete-log signature systems, allowing exposed public keys to be forged.

ClearCrest is proposed as a conservative redesign of that architecture. Its monetary rules remain simple: 50 CCR begin each issuance epoch, rewards halve every 210,000 blocks, one CCR equals 100,000,000 quanta, and total issuance cannot exceed 21,000,000 CCR. The protocol uses a ten-minute target interval and a flat minimum relay fee of 250 quanta. The objective is not to make cryptography invisible; it is to make the cryptographic choice explicit, versioned, and replaceable under defined validation rules.

2. Design Principles

Consensus is deliberately narrow. Every amount is represented as an integer number of quanta and is checked against the maximum supply before it affects state. Lock height and lock time are separate fields. Coinbase transactions include their block height. The difficulty calculation consumes an exact 2,016-block interval. Timestamps are 64-bit values and must satisfy median-time rules. These choices remove ambiguity that can turn edge cases into consensus splits.

Cryptographic agility is a consensus feature, not a wallet preference. Address and signature versions identify the validation algorithm. Version 1 is reserved for ML-DSA-65 and version 2 for SLH-DSA. A node accepts a signature only through a reviewed provider registered for that version. An implementation with no such provider must fail closed; it must not silently substitute an unrelated or home-grown signature scheme.

3. Transactions

A transaction contains a version, type, inputs, outputs, independent lock height and lock time, an opt-in replacement flag, and type-specific data. Inputs reference a prior transaction identifier and output index. Outputs bind an amount to a recipient commitment and may carry an ephemeral public value when a supported privacy construction requires one. Transaction identifiers are the double SHA3-256 hash of one canonical byte serialization.

Transaction
Version | Type | Lock Height | Lock Time | RBF
Inputs: previous identifier, index, signature version, signature, public key
Outputs: amount, recipient commitment, optional ephemeral public value

For ordinary transfers, validation requires distinct existing inputs, valid signatures for the declared version, a non-negative fee of at least 250 quanta, and input value no less than output value plus fee. Every output must lie between zero and the maximum supply. A replacement transaction may displace a transaction only when both spend exactly the same inputs, both opt in to replacement, and the replacement pays a strictly higher fee.

4. Timestamp Server and Proof of Work

Nodes gather valid transactions into a block. A block header commits to the previous block hash, the transaction Merkle root, a compact-filter root, a 64-bit timestamp, target, nonce, and version. The block identifier is the double

SHA3-256 hash of this fixed-width header. A valid proof of work is a header hash no greater than the target. The best chain is selected by accumulated validated work, subject to the full consensus rules.

Targets adjust from an exact 2,016-block timestamp window. The observed interval is clamped to one quarter through four times the expected interval before a new target is calculated. Median-time-past prevents a single timestamp from moving the clock arbitrarily, including at adjustment boundaries. Blocks may serialize to no more than 6 MiB.

5. Network

The network broadcasts inventory and retrieves only unknown data. Frames begin with a ClearCrest magic value, a fixed command field, a bounded payload length, and a truncated double-SHA3 checksum. Supported messages include version negotiation, inventory, block and transaction retrieval, compact blocks, compact filters, and channel alerts. Invalid commands, oversized messages, malformed lengths, and checksum failures are rejected before a payload reaches consensus processing.

Because post-quantum signatures are larger than conventional signatures, block propagation must avoid repeated full-block transmission. A compact block sends the header, selected prefilled transactions, and short transaction identifiers derived from a per-block key. A receiving node reconstructs the remainder from its validated mempool and requests only identifiers it cannot resolve. Short identifiers are transport hints only; final acceptance always rechecks the full transaction identifiers and consensus rules.

6. Incentive and Supply

The first transaction of each valid block is a coinbase transaction. Its height is committed in the transaction data, making its identifier unique across blocks. Its value equals the scheduled subsidy at that height plus fees permitted by the block. The initial subsidy is 5,000,000,000 quanta and halves every 210,000 blocks. No calculation may create more than 2,100,000,000,000 quanta in total. Coinbase outputs mature after 100 completed blocks.

7. Storage, Pruning, and Light Clients

A full validator maintains the unspent transaction output set and applies each block transactionally. Persistent storage indexes block height, block hash, outputs, filters, and channel state. A pruning node may retain the unspent set, chain headers, filters, and a configurable recent block window while discarding older full block payloads after their state effects have been validated.

Each block also commits to a compact membership filter. A light client downloads headers and filters, tests locally for its own address commitments, and requests full data only for probable matches. The filter has false positives but must never create a false negative for an inserted item. This reduces bandwidth without requiring a client to reveal its complete address set to a remote node.

8. Native Payment Channels

A channel locks two participants' funds into a jointly defined balance state. It progresses from opening to open, then either cooperative close or a bounded dispute state, and finally closed. Each accepted off-chain state has a monotonically increasing number. A periodic on-chain anchor commits to a recent state number. A force close whose state number is lower than the latest valid anchor is invalid by consensus.

A unilateral close opens three 144-block challenge phases. The protocol can reserve a challenge bounty of 5 percent, 15 percent, and 20 percent of the closer's allocation across those phases. A revocation proof for a stale state awards the applicable bounty to the challenger. The bounded window prevents an adversary from indefinitely locking a channel while the increasing bounty creates a direct economic reason to monitor it. Watchtowers are optional agents that retain revocation material and submit penalty evidence when authorized.

9. Hash Time Locked Contracts and Atomic Swaps

An HTLC locks value to a payment hash and a timeout height. The designated receiver claims before expiry by revealing a preimage whose SHA3-256 digest equals the committed payment hash. After expiry, an unclaimed output returns to the sender. Timeouts are constrained to a configured range so multi-hop payments and cross-chain constructions have sufficient settlement margin.

An atomic swap follows from compatible hash and time locks. One party locks CCR under a secret hash; the counterparty locks value on another compatible chain under the same hash with a shorter or longer timeout chosen to

preserve a safe refund order. A claim reveals the secret and permits the reciprocal claim. Interoperability remains conditional on the other chain's actual scripting, confirmation, and reorganization rules.

10. Privacy and Key Recovery

ClearCrest addresses encode a key version and a truncated SHA3 commitment with a Bech32m checksum under the cc prefix. Regular addresses remain public recipient commitments. A future stealth-address construction may publish separate scanning and spending material and derive one-time recipients, but it must use a reviewed post-quantum key-establishment design before deployment. It is not sufficient to adapt a classical Diffie-Hellman scheme and call the result quantum safe.

Wallets may offer Shamir secret sharing for disaster recovery. A private secret is split into N shares with a K-share recovery threshold over a finite field. No individual share reveals the original secret. The shares are recovery material, not transaction authority, and should be stored separately with independently reviewed encryption and access controls.

11. Security Considerations and Deployment Gates

The protocol depends on honest majority proof-of-work assumptions, correct node implementations, secure peer-to-peer transport behavior, and a sound implementation of each enabled signature version. Post-quantum standardization does not eliminate implementation risk: key encoding, side-channel behavior, randomness, parameter selection, interoperability, and library lifecycle management require review. An operational deployment also requires reproducible builds, deterministic test vectors, fuzzing, denial-of-service limits, incident response, monitoring, backups, and independent audits.

Accordingly, the reference implementation is not a launch authorization. Before any production network, the project must complete a public specification review, cross-implementation test suite, signed release process, cryptographic provider audit, application security audit, testnet period, and a documented governance process for version activation. No funds should be entrusted to a pre-audit implementation.

12. Conclusion

ClearCrest applies the peer-to-peer cash model to a future in which signature agility is mandatory. Its core proposal is direct and testable: a proof-of-work ledger with bounded integer values, canonical messages, versioned post-quantum verification, efficient relay, prunable validation state, and first-class conditional payments. The design is only as strong as its implementation and review. Its path to usefulness is therefore staged engineering and public verification, not an unsupported claim of finished security.

References.

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